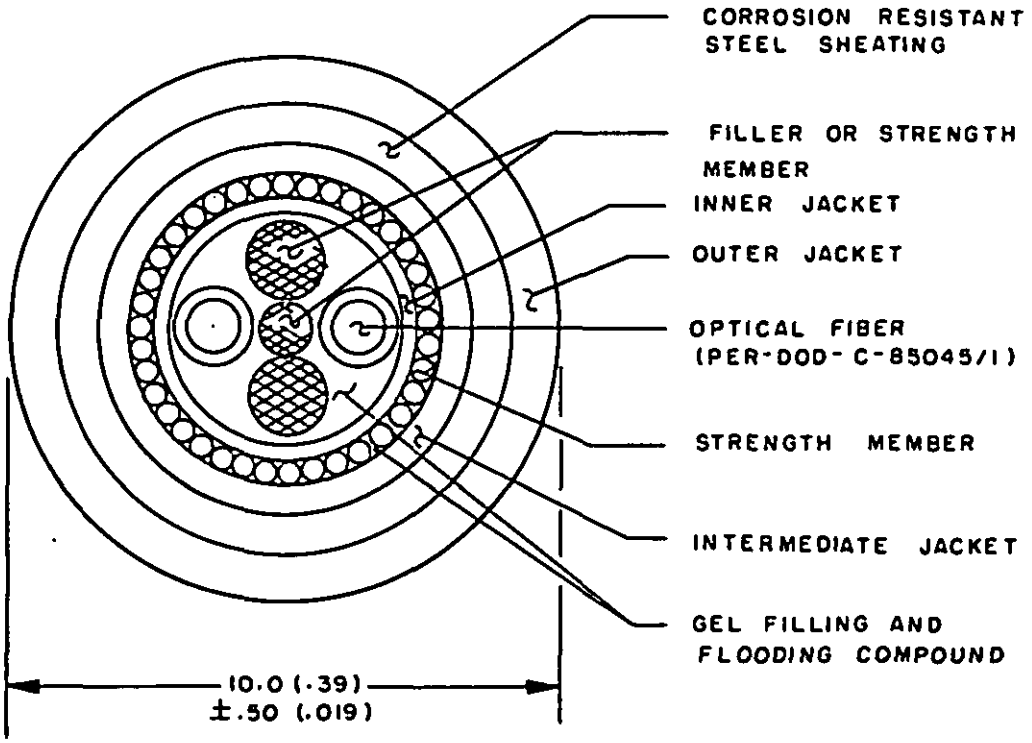
[METRIC]
DOD-C-95045/4A
20 March 1987
SUPERSEDING
DOD-C-85045/4
22 June 1984

MILITARY SPECIFICATION SHEET

CABLES, FIBER OPTICS, HEAVY DUTY, RUGGEDIZED WITH STEEL SHEATHING RODENT PROTECTION, GEL FILLING AND FLOODING COMPOUND (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cables described herein shall consist of this specification and the latest issue of DOD-C-95045.



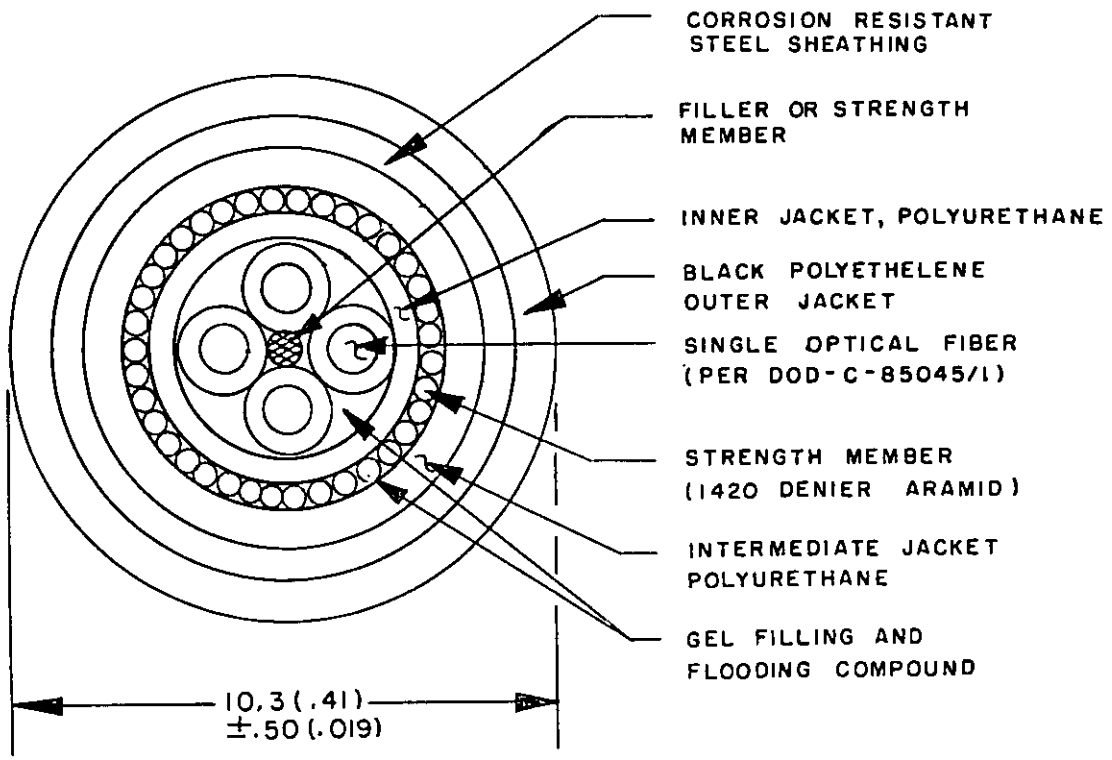
NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are in parentheses.

FIGURE 1. Two fiber cable.

AMSC N/A FSC 6015

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

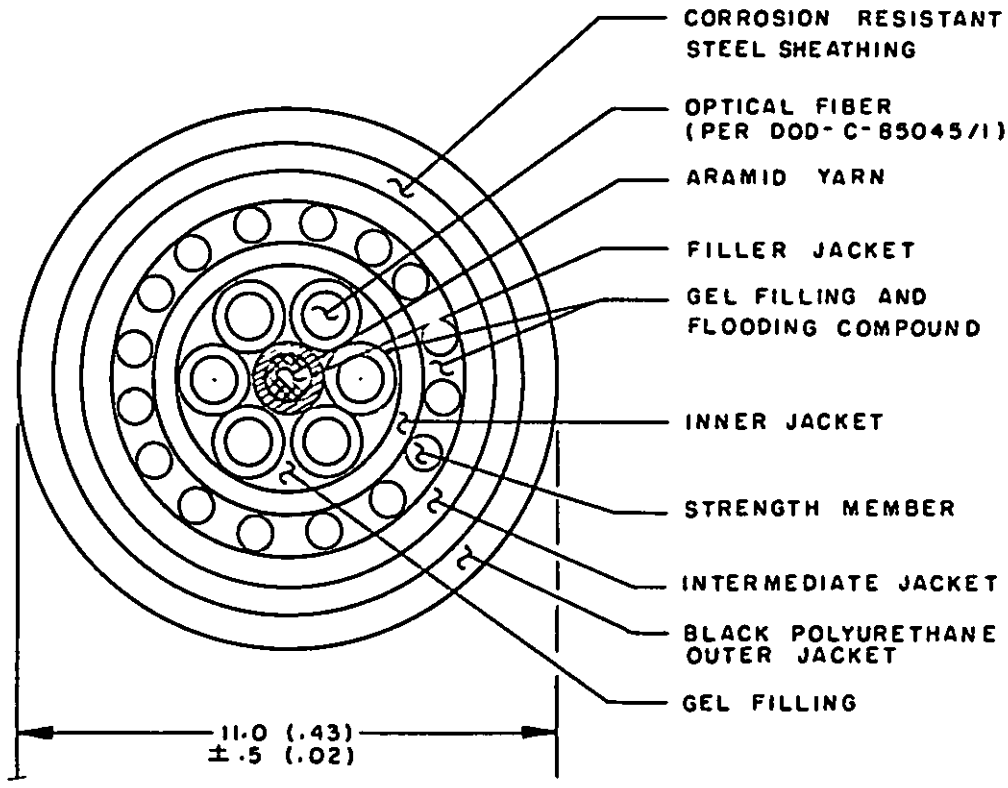
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:06Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions are in millmeters.
2. Inch equivalents are in parentheses.

FIGURE 2. Four fiber cable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:06Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are in parentheses.

FIGURE 3. Six fiber cable.

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:06Z
Check the source to verify that this is the current version before use.

TABLE I. Part numbers.

Part number D85045/4-	Number of fibers	Maximum weight (kg/km)
B2B	2	115
C2B	2	115
B4B	4	120
C4B	4	120
B6B	6	140
C6B	6	140

REQUIREMENTS:

Dimensions and configuration: Fiber cables shall be designed as in figure 1 through 3.

Weight: See table I.

Fiber: Shall be in accordance with DOD-C-85045/1.

Number of fibers: See table I.

Jacket material: Polyurethane or equivalent.

Filler and strength member material: Aramid yarn or equivalent.

Color: Outer cable jacket color shall be black.

Attenuation: Attenuation shall not exceed 3.0 dB/km at 1100 to 1300 nm or 6 dB/km at 850 ±20 nm for each and every fiber in the cable.

Optical fiber bandwidth (fiber transfer function): The optical fiber bandwidth shall meet or exceed 600 MHz-km at 1300 ±20 nm and 400 MHz-km at 850 ±20 nm.

Numerical aperture: 0.20 ±0.02.

Loose tube buffers: Constructed of thermoplastic polymer capable of containing up to five individual fibers.

Temperature cycling: -46°C and +71°C.

Impact: Test shall be performed at 25°C ±2°C, 200 impacts. Test mass - 3 kg.

Crush: Test shall be repeated at +85°C ±2°C. The test load of 1500 newtons shall be applied within 5 seconds; hold time shall be 10 seconds minimum, release time shall be 5 seconds maximum.

Wicking: Procedure II, except that the test duration shall be 168 hours. Maximum dye penetration shall be 5 cm.

Tensile loading: 2,700 newtons minimum.

Storage temperature: The cable shall be stored for 240 hours minimum at -55°C ±2°C and +85°C ±2°C. Following storage at each temperature, the attenuation measured at ambient conditions shall not exceed that specified herein and the exterior surface shall show no cracks or defects.

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:06Z
Check the source to verify that this is the current version before use.

Fluid immersion (DOD-STD-1678, method 8030): After testing, the cable diameter shall not exceed the maximum cable diameter specified. Fluids, temperature and immersion time shall be as specified below.

Fluid	Temperature (C°)	Immersion time (hours)
Lubricating oil MIL-L-23699	48 - 50	20
Hydraulic fluid MIL-H-5605	48 - 50	20
Isopropyl alcohol TT-I-735	20 - 25	168
Turbine fuel MIL-T-5624	20 - 25	168
Automobile gas MIL-G-3056	20 - 25	168

Freezing water immersion (ice crush): The specimen length shall be off reel and loosely coiled. Except for access ends, the test specimen shall be completely immersed in water; the vessel containing the water need not be sealed. The attenuation measured in step 9 shall not exceed that specified herein.

Jacket opacity: Shall be verified at 1300 nm ±50 nm.

Parameter applicability: See table II.

Gel filling and flooding compound compatibility: The gel filling and flooding compound shall be compatible with the fiber material.

TABLE II. Parameter applicability. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Group I					
Visual and mechanical	6 units 1 km each	X	X		
Attenuation rate	6 units 1 km each	X	X		
Optical fiber bandwidth	6 units 1 km each	X	X		
Crosstalk	6 units 1 km each	X	X		
Numerical aperture	6 units 2 m each (1 each from 6 cables)	X	X		
Group II					
Temperature cycling	6 units 1 km each (3 on reel, 3 off reel)	X		X	
Humidity	2 units 1 km each	X		X	
Storage temperature	1 units 1 km each (1 on reel, 1 off reel)	X		X	
Barometric pressure (reduced)	1 unit 1 km	X		X	
Tensile loading	6 units 150 m each (2 each from 3 cables)	X		X	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:06Z
Check the source to verify that this is the current version before use.

TABLE II. Parameter applicability - Continued. 1/ 2/

Inspection	Specimen quantity and length	Qualification	Quality assurance group A	Quality assurance group B	Quality assurance group C
Freezing water immersion (ice crush)	2 units 30 m each (1 each from 2 cables)	X		X	
Group III (Military environments)					
Low temperature flexibility (cold bend)	3 units 8 m each (1 each from 3 cables)	X			X
Crush	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Cable twist-bending	6 units 5 m each (3 each from 2 cables, 2 units for each temperature)	X			X
Impact	9 units 5 m each (3 each from 3 cables)	X			X
Jacket self-adhesion or blocking	2 units >5 m each (1 each from 2 cables)	X			X
Wicking	2 units 2 m each (1 each from 2 cables)	X			X
Cable jacket opacity	3 units 2 m each (1 each from 2 cables)	X			X
Fluid immersion	2 units (for each test fluid required) 2 m each (1 each from 2 cables)	X			X
Knot	6 units 5 m each (3 each from 2 cables)	X			X
Ultraviolet radiation	3 units 2 m each (1 each from 3 cables)				
Cable element removability	3 units 2 m each (1 each from 3 cables)	X			X
Durability of identification marking	3 units 2 m each (1 each from 3 cables)	X			X

1/ Tolerance on 1 km lengths is ±2% provided results are normalized to 1 km.

2/ Tolerance on shorter lengths is ±10%.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:06Z
Check the source to verify that this is the current version before use.

Supersession data: See table III.

TABLE III. Supersession data.

Superseding part number DOD-C-85045/4-	Superseded part number DOD-C-85045/4-
B2B	2C
C23	2D
B4B	4C
C4B	4D
B6B	6C
C6B	6D

Revision letters are not used to denote changes due the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, SC
Navy - AS
Air Force - 11
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6015-0013-3)

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:06Z
Check the source to verify that this is the current version before use.